

# PROVISIONAL PATENT APPLICATION

## **Configuration-Ledger Molecular Design and Control Systems: Convergence-Certifying Molecular Processor, Spectroscopic Measurement Controller, Reaction and Synthesis Controller, Materials Fabrication Controller, and Diagnostic or Therapeutic Energy-Delivery Controller, and Methods of Operating Them**

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*This specification is complete and self-contained. No companion application is relied upon; no material essential to written description, enablement, or the claims is incorporated from another document. Every example identified as prophetic is prospective; no experiment, measured result, clinical result, manufactured material, or controlled reaction is represented as achieved.*

*Portfolio prophetic-example statement: All examples identified as prophetic are prospective; no prophetic example is represented as an experiment performed or a result achieved.*

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### **FIELD**

The invention relates to computer-controlled molecular and materials systems and, more particularly, to processors that construct a finite configuration ledger from endpoint census records, overlap records, and primitive pair-posting records; determine stationary molecular configurations and permitted configuration changes; issue convergence, chart-invariance, duplicate-posting, and non-closure certificates; and control physical measurement, synthesis, fabrication, catalyst-screening, energy-conversion, diagnostic, or therapeutic apparatus from the certified result.

### **BACKGROUND**

A molecular result may be numerically converged within a finite representation, bounded relative to a specified larger model, or supported by measurements in an operating envelope. These are different assurances. A downstream instrument receiving only a target geometry, spectrum, pathway, pulse sequence, or material composition cannot determine which assurance applies, which effects were omitted, or whether the stated uncertainty is adequate for its next physical command.

Existing molecular solvers also commonly separate electronic-structure calculation, conformer generation, transition-path calculation, spectroscopy, pulse design, and apparatus control into different products. Translation between them may lose the integer census, symmetry, provenance, uncertainty, and failure information that determined the candidate. A failed closure may be returned as a poorly converged number rather than as a distinct non-closure output.

There remains a need for a processor and controlled apparatus in which: (a) the molecular representation is built from finite, auditable records; (b) duplicated filing is removed by an overlap book rather than silently counted twice; (c) primitive one-entry and pair-entry records uniquely assemble a many-entry configuration book; (d) enlargement continues by a predetermined rule rather than molecule-specific repair; (e) the result carries convergence and validity certificates into the physical controller; and (f) invalid or non-closing results inhibit physical action.

Applicant uses a physical ontology in which persistent molecular organization is a finite census of closed circulations and two-ended, co-signed relations. Conventional terms such as “orbital,” “electron,” “Hamiltonian,” “wavefunction,” and “energy level,” where quoted or used in a translation embodiment, identify customary numerical interfaces and do not limit the disclosed apparatus to an entity ontology historically associated with those terms. The claimed machines operate on recited records, measurements, control states, and physical outputs.

## SUMMARY

A configuration-ledger processor receives endpoint records describing participating elemental or material seats, an integer occupancy census, a finite set of admissible circulation charts, and apparatus constraints. It forms (i) a Gram or overlap book  $G$  recording duplicated filing among charts and (ii) an action-rate or cost book  $B$  assembled from one-entry records and a primitive four-index pair-posting tensor. Stationary records satisfy a generalized ledger relation  $Bc = \beta Gc$  or an equivalent discrete no-lowering test. Configurations having a zero oriented Gram volume, non-integer endpoint census, failed two-ended posting, or failed closure are rejected.

The processor enlarges the admissible chart by a predetermined, element-independent closure rule. One preferred rule forms a block action-closure ladder  $K_q = \text{span}\{C, (G^{-1} B)C, \dots, (G^{-1} B)^q C\}$ , using the full multi-endpoint book including cross-endpoint postings, and reconciles Gram-duplicated directions at each rung. Registered changes in selected roots, moments, tensor invariants, and output coordinates on two successive representation levels are convergence checks. An invariant-projector closure test or equivalent spectral-reachability construction additionally tests whether the selected space is closed under the declared generator. Iteration counts are recorded for reproducibility, not required to coincide under chart changes. Alternative deterministic ladders may use predetermined radial dilation, angular-harmonic completion, sparse-grid refinement, finite-element subdivision, or other representation enlargement fixed before comparison with the target output.

A certification engine attaches to each result: the chart and basis identifiers; primitive-record provenance; census and duplicate-rejection status; convergence history; uncertainty; validity envelope; held-out or reference-control status; and one of valid, degraded, invalid, or non-closing states. A physical controller is permitted to actuate only an output whose certificate satisfies stored rules. Thus the processor is not merely a calculator; it is a safety and provenance interlock for laboratory and manufacturing apparatus.

The certification engine separately identifies numerical convergence, a conditional bound relative to a declared reference model, exact reduction relative to that model when supported, and empirical validation. An omitted-direction assessor tests coupling to a declared complement and does not infer completeness from full rank in the supplied chart. An output-bound engine propagates supported bounds to a requested observable or apparatus command. Unknown error contributions remain unknown; they are not converted into zero error. Exact-reduction and error-enclosure embodiments described below are conditional operating modes, not assertions that every molecule admits such a reduction or that universal physical exactness or computational superiority has been achieved.

Product branches include: (I1) a configuration-ledger molecular processor; (I2) a spectroscopic measurement and inverse-design controller; (I3) a reaction, synthesis, or molecular-assembly controller; (I4) a materials design and fabrication controller; (I5) a diagnostic or therapeutic energy-delivery controller; and (I6) a catalyst or energy-conversion screening controller. Each branch may operate in a direct native mode, a calibrated mode using measured primitive pair-posting records, or a translation mode using conventional integral machinery to populate the same disclosed records.

The disclosure does not claim a mathematical relation in the abstract. It claims particular computing systems, data structures, certification logic, apparatus interfaces, interlocks, and closed-loop methods that transform measured or specified endpoint records into a certified physical control program and then measure the resulting physical system.

## **BRIEF DESCRIPTION OF THE DRAWINGS**

- **FIG. 1** — Configuration-ledger processor and controlled apparatus.
- **FIG. 2** — Native data structures: endpoint census, Gram book, primitive pair-posting tensor, and assembled configuration book.
- **FIG. 3** — Deterministic re-filing and convergence-certification ladder.
- **FIG. 4** — Validity state machine and physical-action interlock.
- **FIG. 5** — Spectroscopic measurement and inverse-design controller.
- **FIG. 6** — Reaction, synthesis, and molecular-assembly controller.
- **FIG. 7** — Materials design and fabrication controller.
- **FIG. 8** — Diagnostic or therapeutic energy-delivery controller.
- **FIG. 9** — Catalyst and energy-conversion screening controller.
- **FIG. 10** — Factory qualification, digital record, and layered scope.
- **FIG. 11** — Scope-aware certification, omitted-direction assessment, output-bound propagation, and adaptive apparatus-command gate.

**DEFINITIONS (the applicant acts as his own lexicographer; these definitions govern the claims)**

1. **Endpoint or seat** means a persistent participating center represented by an elemental, atomic, molecular-fragment, lattice-site, defect-site, surface-site, or apparatus-coupling record. It does not require a point particle.
2. **Census entry** means a persistent whole-entry record including an endpoint identifier, an integer occupancy, an orientation or sense identifier where applicable, a radial or reach chart, a phase or registry field, and provenance. "Integer" includes zero and signed whole values where the record type permits them.
3. **Chart** means a finite numerical description used to post one real circulation or relation. Multiple charts may describe overlapping portions of the same entry. A chart is not itself the physical entry.
4. **Gram book or overlap book** means a stored matrix or equivalent data structure  $G$  whose entries quantify common filing among charts and whose null or near-null directions identify duplicated descriptions. It is used to prevent the same physical entry from being counted more than once.
5. **Primitive pair-posting tensor** means a stored four-index record  $P_{abcd}$ , or a symmetry-reduced equivalent, giving a signed pair entry between two chart pairs before many-entry assembly. The tensor may be generated directly from a native circulation ledger, measured or calibrated, or produced through a conventional numerical translation.
6. **Configuration book** means a matrix, sparse operator, tensor network, or equivalent finite data structure assembled from one-entry postings and primitive pair postings for a whole admissible configuration. In a preferred exterior-cofactor assembly, pair-only additivity and oriented duplicate rejection determine the signs and cofactors.
7. **Stationary configuration** means a configuration for which no admissible whole-entry re-filing lowers the selected action-rate or cost ledger while endpoint census, closure, linking, two-ended posting, and apparatus constraints remain satisfied. The generalized relation  $B_c = \beta G_c$  is one numerical representation of that discrete condition.
8. **Re-filing** means redistributing the chart description of a persistent whole census without creating a fractional persistent entry. A re-filing direction need not itself be an occupied persistent entry.
9. **Deterministic closure ladder** means a representation-enlargement sequence fixed without using the target molecule's observed geometry, spectrum, rate, or product yield. The same rule is applied to every endpoint of a declared class.
10. **Closure certificate** means a machine-readable record stating whether endpoint census, two-ended posting, Gram duplicate rejection, integer admissibility, and applicable geometric or phase closure tests pass.

11. **Convergence certificate** means a machine-readable record containing at least two successive representation levels, changes in predetermined invariants or output quantities, stopping thresholds, and a pass or fail state. Numerical convergence alone does not certify the completeness or physical accuracy of the represented model.
12. **Chart-invariance certificate** means a machine-readable result of rerunning or transforming a calculation under a nonsingular chart change and comparing invariant roots or outputs to a stored tolerance.
13. **Non-closing output** means an affirmative result distinct from numerical failure. It identifies the declared search space and failed rule or witness. Exhaustive exclusion or a mathematical infeasibility witness may establish nonexistence within that scope; termination of an incomplete search, insufficient precision, or exhaustion of resources does not establish that a physical configuration cannot exist.
14. **Validity envelope** means stored bounds on inputs, representation error, apparatus state, calibration age, uncertainty, environmental conditions, and output sensitivity within which a control result may be used.
15. **Control program** means machine-readable setpoints, timing, spatial coordinates, compositions, source phases, amplitudes, frequencies, temperature or pressure schedules, flow rates, deposition commands, or stop conditions deliverable to physical apparatus.
16. **Moving-eye correction** means a recomputation or perturbative update in which endpoint positions, endpoint inertias, registry, or motion participate in the same ledger rather than remaining fixed parameters.
17. **Direct native mode** means operation in which the primitive records are generated from closed-circulation census and signed pair-book rules. **Calibrated mode** means operation using primitive records derived from measurements with uncertainty and provenance. **Translation mode** means operation using conventional numerical machinery to populate the disclosed Gram and pair-posting records. Use of translation mode does not redefine the invention's native ontology.
18. **Preparation record** means a record of actual initial conditions relevant to a physical event, including composition, geometry, temperature, pressure, motion, source registry, pulse history, boundary condition, and uncertainty. A reaction-time or yield output is not certified without a preparation record adequate for the stated result.
19. **Certificate scope record** means a machine-readable declaration of the generator or model, representation, configuration treatment, retained and omitted effects, state or spectral sector, requested observable, parameter region, preparation where applicable, units, numerical precision, and proof or measurement provenance to which a certificate applies. A status of VALID is always relative to this record and an apparatus acceptance policy.

20. **Omitted-direction assessment** means evaluation of coupling from a retained subspace to a specified complementary subspace or unresolved tail. A finite enlargement is a probe of that enlargement only unless a stated argument bounds or excludes the entire remaining complement.
21. **Exact-reduction certificate** means a record supported by an analytic identity or verified proof that a retained subspace is invariant under a declared reference generator, contains the state or prepared-state sector of interest, and preserves the requested output under the stated conditions. It does not assert that the reference generator includes all physical effects or that a floating-point result is an exact real number.
22. **Output enclosure** means an interval or other set proved, under identified assumptions, to contain the requested reference-model output. An empirical error estimate or statistical confidence interval is identified separately and is not labeled a deterministic enclosure.
23. **Reference generator** means the declared action-rate operator against which a retained calculation is assessed. It may be a larger finite configuration book or a specified full model with a defined domain. A converged self-consistent mean book remains a distinct approximation to a joint-configuration generator unless their equivalence is independently established.

## DETAILED DESCRIPTION

### SYSTEM ARCHITECTURE (FIGS. 1–4)

Referring to FIG. 1, system 100 comprises input interface 110, endpoint-census store 120, primitive-record store 130, ledger processor 140, certificate engine 150, control-program generator 160, apparatus interface 170, physical apparatus 180, measurement return 190, and provenance store 195. Input may be a proposed composition, measured specimen record, target spectrum, desired product, target material property, anatomical target, or apparatus constraint.

Endpoint-census store 120 holds records 210 shown in FIG. 2. Gram constructor 220 forms  $G$  from chart overlaps. Pair engine 230 forms or retrieves  $P_{abcd}$ . Configuration assembler 240 builds  $B_{config}$  by adding one-entry contributions and the oriented cofactors of every primitive pair contribution. Duplicate detector 250 rejects a repeated orientation-and-sense slot when its oriented Gram volume is zero or below a stored numerical tolerance.

In a preferred two-entry implementation, the processor solves  $B_c = \beta G_c$  after Gram whitening or a numerically equivalent factorization. In a many-entry implementation, the processor enumerates or generates admissible integer configurations; computes each configuration's oriented Gram volume; excludes zero-volume configurations; assembles the configuration Gram book and configuration action-rate book from primitive postings; and solves the resulting finite generalized ledger. Sparse, iterative, tensor-network, or block-diagonal implementations may be used without changing the disclosed records.

Referring to FIG. 3, seed chart  $C$  is expanded at ladder rung  $q$  by applying the reconciled primitive action map. Gram-null residuals are discarded as duplicate descriptions. At each rung the processor records stationary roots, radial moments, angular tensor invariants, geometry, and one or more application

outputs. A convergence comparator tests the registered thresholds. The processor does not select a rung because it better matches an observed target.

The convergence certificate preferably includes: software and instrument version; endpoint-record hashes; chart identifiers; quadrature or discretization settings; primitive tensor checksum; rung sequence; root tracking; invariant drift; geometry drift; Hessian drift; uncertainty budget; certificate scope record; and adverse-control results. Chart tests include nonsingular transformations and, separately, redundant embeddings with Gram-null reconciliation. Projectors are compared after transport into a common metric and ordering, not by comparing raw coordinate entries or iteration counters.

Referring to FIG. 4, certificate engine 150 sets a state of QUALIFYING, VALID, DEGRADED, INVALID, or NON-CLOSING. Control-program generator 160 may issue physical commands only in VALID state, may issue bounded diagnostic commands in DEGRADED state, and inhibits energetic, irreversible, clinical, or manufacturing action in INVALID or NON-CLOSING states. A human override, if provided, is logged and cannot erase the failed certificate.

Direct native mode forms primitive range postings by summing the signed pair ledger on actual finite circulation entries. Completed isotropic census contributions may be analytically reduced; unfinished directional census contributions remain unaveraged. The outer directional census may include radial dilation and angular handles generated by the deterministic closure ladder. No observed molecular dimension is used to choose the ladder.

In calibrated mode, pair records are measured using reference dimers, fragments, sites, or apparatus-coupled specimens. Each record includes uncertainty, environmental conditions, preparation, calibration chain, and validity envelope. In translation mode, conventional integral evaluation may supply overlap, one-entry, and pair-entry arrays. If a conventional stationary solver also supplies a self-consistent mean-field or restricted-configuration result, its approximation is separately recorded; it is not described as integral evaluation alone. The subsequent duplicate rejection, exterior assembly where applicable, certificates, apparatus interlock, and closed-loop control remain as disclosed.

## **SCOPE-AWARE REDUCTION AND ERROR CERTIFICATION (FIG. 11)**

Referring to FIG. 11, certificate engine 150 includes scope registry 1110, invariant-space engine 1120, complement assessor 1130, bound evaluator 1140, output propagator 1150, adaptive refinement scheduler 1160, and command-policy gate 1170. The following constructions are numerical representations of co-signed circulation records; they do not introduce another physical substance. They specify conditional machine operations independently of whether an arbitrary molecular model admits a compact exact reduction.

**Reference and assurance types.** Scope registry 1110 stores separate flags for NUMERICALLY-CONVERGED, MODEL-BOUNDED, and EXACT-REDUCTION, together with an empirical-validation record. None is silently promoted into another. The record identifies whether the calculation uses a one-entry, self-consistent mean-field, selected-configuration, complete finite-configuration, fixed-endpoint, moving-endpoint, closed-boundary, or open-boundary model. A finite complete-configuration solution is

exact only for that finite model to its arithmetic tolerance. An observed agreement is evidence about a test case, not a proof of completeness. The controller may accept an explicitly calibrated engineering estimate where its policy permits one; a command requiring a deterministic enclosure cannot be authorized by an estimate of a different type.

**Invariant reachable-space implementation.** In a finite Hermitian implementation, true Gram-null directions are removed and the positive Gram book is factored. Set  $A = G^{(-1/2)} B G^{(-1/2)}$  and transform the occupied seed to  $C\_tilde = G^{(1/2)} C$ . Near-null directions discarded at nonzero tolerance are recorded as numerical truncations. The invariant-space engine forms  $\text{span}\{C\_tilde, A C\_tilde, \dots, A^q C\_tilde\}$ . Equivalently, for spectral projectors  $Pi\_lambda$  of  $A$ , it forms the direct sum of  $\text{range}(Pi\_lambda C\_tilde)$  over eigenvalues having a nonzero projected seed. For a degenerate eigenvalue only the seed-generated part is necessarily reachable; the whole degenerate eigenspace is not automatically occupied or required. Exact spectral reachability and exact Krylov closure coincide for this specified finite Hermitian generator.

Let  $U$  contain orthonormal columns for the retained space,  $P = U U^*$ , and  $Q = I - P$ , where  $*$  denotes the adjoint and  $I$  the identity. Here  $P$  denotes a projector, not the four-index pair-posting record  $P\_abcd$ . The invariant-space engine records rank, seed-containment residual, projector-transport defect, and  $\eta = \|Q A\_ref P\|_2$  for the declared reference generator  $A\_ref$ . Residuals have action-rate units; normalized residuals use a stored nonzero reference scale. A projector is not complete merely because its rank equals the dimension of its supplied chart:  $Q$  is then zero inside that chart by construction. The complement assessor instead uses a declared enlarged chart or a justified analytic tail estimate. Repeated enlargement may establish numerical stability without proving a bound on an unrepresented tail.

For a nonlinear self-consistent generator, the engine records the fixed point and generator-update residual and repeats the closure assessment after the update. Closure of one frozen linear map does not establish stationarity against all joint-configuration changes, uniqueness of a fixed point, or a global molecular minimum. An omitted lower state that is uncoupled from the seed can invalidate a ground-state claim while leaving  $\eta$  zero. A ground-state certificate therefore also needs an ordering or lower-bound witness for the omitted sector, or must be explicitly restricted to the prepared or symmetry sector represented by the seed.

**Residuals and conditional enclosures.** For a unit Ritz vector  $y$  and scalar  $\lambda\_hat$ , the bound evaluator computes  $r = A\_ref y - \lambda\_hat y$ . For a finite Hermitian reference, or a self-adjoint reference with  $y$  in its domain, a verified bound  $\rho \geq \|r\|_2$  places at least one spectral value in  $[\lambda\_hat - \rho, \lambda\_hat + \rho]$ . A verified operator-evaluation error  $\delta\_A$  adds conservatively to  $\rho$ . This existence statement is not a certificate that the interval contains a particular ground, excited, or transition state; state isolation, ordering, or overlap evidence must identify the requested branch. Interval arithmetic with outward rounding or another verified arithmetic enclosure may be used to bound integral evaluation, factorization, residuals, and eigenvalue intervals. An ordinary floating-point residual without its evaluation error remains a diagnostic, not a rigorous enclosure.

An optional complementary-block implementation forms  $A_{PP}$ ,  $A_{PQ}$ ,  $A_{QP}$ , and  $A_{QQ}$  from  $A_{ref}$ . If a verified distance  $d > 0$  separates a trial spectral interval from the spectrum of  $A_{QQ}$ , elimination of the complementary block gives the effective retained map  $A_{PP} + A_{PQ}(\lambda I - A_{QQ})^{-1}A_{QP}$ . With a verified  $\eta \geq \|A_{QP}\|_2$ , the norm of this correction is at most  $\eta^2/d$ . Interval root inclusion, together with branch-isolation tests, converts that correction into an enclosure. The norm estimate alone is not substituted for those tests. If  $d$  cannot be bounded away from zero, the engine enlarges the retained space or declines the bound. For a specified infinite reference, domain and unresolved-tail estimates are required in addition to any finite-block calculation.

The bound record separately accounts for radial/angular representation, quadrature or integral translation, arithmetic and conditioning, joint-configuration truncation, endpoint-motion approximation, boundary treatment, and calibrated-input uncertainty. Each contribution is marked bounded, estimated, not applicable, or unknown and carries units and provenance. A conservative total is formed only from compatible bounds propagated to the same output; unknown contributions cannot be entered as zero. A model-relative bound does not include discrepancy between the model and an actual specimen unless that discrepancy is independently characterized over the declared envelope.

**Observable-specific propagation.** A stationary-ledger-value enclosure does not by itself enclose geometry, modes, barriers, or event times. For a locally differentiable geometry branch, the processor may certify a stationary point  $q_{star}$  in a declared convex basin after removing rigid motions and other zero modes. If the exact reference objective in that basin has Hessian bounded below by  $m I$  with  $m > 0$ , the computed gradient norm is at most  $g$ , and gradient-evaluation error is at most  $e_g$  uniformly as required, then  $\|q_{hat} - q_{star}\| \leq (g + e_g)/m$ . The certificate records coordinate units, basin, existence and curvature witnesses. It is a local-basin statement, not global conformer uniqueness. Without those witnesses, geometry drift remains a convergence estimate.

A mode enclosure uses bounds on the mass- or inertia-reconciled Hessian and positive inertia, then maps a verified nonnegative squared-rate interval through the square root. Negative or sign-indeterminate intervals are reported as unstable or unresolved rather than as real mode rates. A barrier enclosure subtracts separately identified saddle and reactant enclosures and carries preparation and branch qualifications. Statistical event-rate statements additionally retain their distributional assumptions. These output-specific operations prevent an energy-like ledger residual from being misused as an apparatus timing, spatial, or selectivity guarantee.

**Motion and adaptive operation.** A projector that varies with endpoint position or time requires differentiation of the metric and chart as well as the ledger. For evolution  $i \hbar \dot{z} = A_{ref}(t) z$ , retention in a moving space requires the outside component  $Q A_{ref} P - i \hbar Q \dot{P} P$  to vanish, not  $Q A_{ref} P$  alone. The engine records the time interval, initial-state containment, boundary condition, and the omitted geometric coupling. Geometry derivatives include chart and Gram derivatives; degeneracies, rank changes, branch crossings, or unresolved outward transfer trigger a new certificate. Static closure is not a certificate of full moving-endpoint or radiative dynamics.

Adaptive refinement scheduler 1160 receives an apparatus tolerance and compute budget. It selects among predetermined radial, angular, joint-configuration, quadrature, arithmetic, or motion refinements using the failing certificate component. In one implementation it ranks eligible actions by the estimated decrease of the dominant bound per unit of measured execution time, with a predetermined tie-break rule, and recomputes the certificate after each action. The ranking estimate never substitutes for the verified acceptance bound. Reusable primitive blocks and projectors may be cached under geometry, scope, and source hashes, but a changed generator is retested before reuse. On budget exhaustion it returns an unresolved or degraded result rather than weakening the requested tolerance. No observed target is used to select added directions. A claimed reduction in execution time or memory would require a separately recorded comparison at the same requested observable, accuracy, model scope, and hardware; no such performance result is asserted here.

Command-policy gate 1170 verifies the certificate type, scope hash, observable interval, validity region, and expiration against the command requirements. It permits a process window, scan band, or placement command only when the applicable interval, including equipment uncertainty and guard margin, lies inside the authorized envelope. A pending run, unknown omitted-space contribution, changed calibration, or expired proof may allow a bounded diagnostic acquisition but cannot authorize a command demanding the missing assurance.

## **11 — CONFIGURATION-LEDGER MOLECULAR PROCESSOR**

Processor 100 accepts a composition and endpoint census and produces one or more ranked stationary configurations, absolute or normalized dimensions, curvature records, registry classes, transition saddles, and permitted whole-entry moves. It also produces failure witnesses including census mismatch, impossible Gram rank, duplicate slot, nonpositive weighted Gram book, unclosed phase registry, unconverged ladder, or output outside a validity envelope.

A geometry engine derives equivalent-domain directions by minimizing the directional overlap debit  $\text{tr}(G^2)$  subject to closure  $G1=0$  and unit diagonal. Unequal positive radial postings  $w_i$  are handled by a weighted Gram book  $H$  having diagonal  $w_i^2$  and closure  $H1=0$ . The geometry engine reports a configuration only where the resulting Gram book is positive semidefinite to tolerance and embeddable in the declared dimension.

A Hessian engine perturbs certified stationary records in declared radial, angular, torsional, or ring coordinates, rebuilds the ledger at each perturbation, and returns finite-difference or analytic curvatures. A rate engine combines a curvature with an endpoint or collective inertia record and reports a phase-rate, mode-rate, or spectral coordinate with propagated uncertainty. A moving-eye engine repeats the calculation with finite endpoint motion and reports the order and magnitude of the correction.

## **12 — SPECTROSCOPIC MEASUREMENT AND INVERSE-DESIGN CONTROLLER (FIG. 5)**

Controller 500 couples processor 100 to source 510, specimen holder 520, environmental control 530, detector 540, timing/reference subsystem 550, and safety interlock 560. The processor predicts or classifies permitted registry changes and spectral coordinates; the controller selects a bounded scan or pulse program; the detector returns amplitude, phase, timing, and environmental witnesses; and update engine 570 modifies only declared primitive records or preparation records, preserving original provenance.

An inverse-design mode receives a target spectral response and searches compositions or configurations whose certified ledger outputs fall within a target band. Candidates with failed convergence or excessive sensitivity are rejected before synthesis. A measurement mode distinguishes a structural mismatch from a preparation mismatch by separately reporting stationary-book and preparation-record residuals.

## **13 — REACTION, SYNTHESIS, AND MOLECULAR-ASSEMBLY CONTROLLER (FIG. 6)**

Apparatus 600 includes reaction vessel or assembly stage 610; controllable sources 620 selected from thermal, pressure, flow, acoustic, optical, electrical, magnetic, mechanical, or chemical-potential sources; sensors 630; product analyzer 640; and controller 650. The processor constructs reactant, intermediate, saddle, and product configuration books from the same primitive records and identifies a whole-entry path. A path requiring fractional persistent census or a duplicated slot is rejected.

Controller 650 generates a control program specifying source registry, sequence, amplitude, duration, temperature, pressure, flow, and stop rules inside equipment limits. A candidate event time or yield is certified only with an adequate preparation record. Closed-loop operation compares product-analyzer output with the certified target, updates the preparation model or calibrated pair records within stored bounds, and never silently alters the native ledger rule.

Molecular-assembly embodiments include automated flow chemistry, catalyst selection, surface assembly, molecular-beam or trap assembly, polymerization, crystal growth, and biomolecular binding assays. A null output may command no reaction, a different preparation, or safe shutdown.

## **14 — MATERIALS DESIGN AND FABRICATION CONTROLLER (FIG. 7)**

Apparatus 700 receives a target property envelope and generates candidate finite clusters, lattices, surfaces, interfaces, defects, dopants, or layered structures. Ledger processor 140 ranks candidates using certified stationary dimensions, curvatures, registry classes, and sensitivity to perturbation. Fabrication interface 710 converts a selected candidate into deposition, lithography, implantation, annealing, mixing, printing, machining, or crystal-growth commands for tool 720.

Metrology return 730 measures composition, structure, response, and defects. Reconciliation engine 740 compares the measured product with the candidate record, updates a calibrated primitive-record branch where authorized, and preserves a separate direct-native branch. The system rejects a candidate where

fabrication tolerances exceed its certified stability margin.

## **I5 — DIAGNOSTIC OR THERAPEUTIC ENERGY-DELIVERY CONTROLLER (FIG. 8)**

System 800 comprises subject or specimen model 810, target-region record 820, source array 830, sensors 840, dose or exposure monitor 850, controller 860, and independent safety interlock 870. The ledger processor identifies target and off-target configurations or registry bands and generates a spatially and temporally bounded source program. Sources may include acoustic, optical, thermal, electrical, magnetic, mechanical, or particle-beam apparatus as conventional equipment labels.

Before delivery, controller 860 requires identity, calibration, convergence, uncertainty, motion, and exposure-limit certificates. During delivery it receives motion and response updates, recomputes or selects a certified control segment, and terminates delivery on certificate failure. No clinical efficacy is asserted. Prophetic uses include selective sensing, contrast generation, ablation, stimulation, drug release, bond-selective processing, and treatment planning, subject to independent safety and regulatory validation.

## **I6 — CATALYST AND ENERGY-CONVERSION SCREENING CONTROLLER (FIG. 9)**

System 900 enumerates catalyst sites, adsorbate configurations, reaction networks, or energy-conversion materials; assembles certified configuration books; and selects candidates based on stationary closure, barrier, curvature, registry, robustness, and apparatus constraints. Robotic test station 910 prepares candidates and measures yield, selectivity, durability, or conversion output. Active-learning controller 920 may select the next experiment using certificate-aware acquisition that penalizes unconverged or extrapolative candidates.

Energy-conversion embodiments include catalytic, electrochemical, photochemical, thermochemical, piezoelectric, magnetocaloric, and phase-change systems. These customary equipment labels do not alter the primitive ledger architecture. The controller reports measured performance separately from native or translated calculated quantities.

## **QUALIFICATION AND CONTROLS (FIG. 10)**

Factory qualification comprises: Q1 record-schema and checksum tests; Q2 Gram duplicate and near-duplicate tests; Q3 chart rotation, reorder, rescaling, and overcomplete-chart tests; Q4 deterministic-ladder convergence tests; Q5 separated-entry and impossible-closure controls; Q6 reference-system holdouts not used to choose the representation; Q7 apparatus dry-run and safe-state tests; Q8 measurement provenance and uncertainty tests; and Q9 closed-loop rollback tests.

A shipped system stores the tested software version, primitive-record library version, apparatus calibration version, allowed endpoint classes, threshold set, reference results, and expiration or recalibration conditions. Results produced outside those bounds are marked DEGRADED or INVALID and cannot command irreversible action unless separately qualified.

The system may expose layered outputs: L0 raw sensor and primitive records; L1 reconciled one-entry and pair-entry books; L2 certified stationary configurations and permitted moves; L3 derived geometry, curvature, spectrum, barrier, or sensitivity; and L4 apparatus control program and measured result. Passing one layer does not imply passage of a higher layer. Scope-aware qualification additionally tests false full-rank completeness, an uncoupled omitted lower state, a vanishing spectral gap, a moving projector with nonzero geometric coupling, incompatible or unknown error contributions, and rejection of a numerically converged certificate for a command requiring a model-bounded observable.

## PROPHETIC EXAMPLES

**P1 — convergence-certified molecular design.** A user supplies an elemental census and a target geometry envelope without supplying an observed geometry for the candidate. Illustrative acceptance settings require the last two changes in registered dimensions and tensor invariants to be below 0.5% and 1%, respectively, and chart-transformed invariant roots to agree within  $10^{-8}$  in declared units. If these checks and the scoped closure checks pass, the processor may issue NUMERICALLY-CONVERGED for the stated model and export coordinates for an authorized use. It will not issue a model-bounded geometry certificate without the additional output-bound evidence. Failed searches and numerical limits are distinguished from demonstrated non-closure.

**P2 — spectroscopic controller.** A source scan is restricted to a certified mode band plus guard intervals. Drive-off, detuned, reference-specimen, and source-reversal controls are randomized. A detected response is assigned to the candidate only when phase, amplitude, environment, and calibration witnesses pass. The example is prospective and no response is represented as measured.

**P3 — reaction controller.** For a specified reactant preparation, the processor identifies a pair-only sequence whose intermediate configuration remains integer and duplicate-free. The apparatus executes a bounded pulse and temperature sequence, analyzes products after each block, and stops on off-target product, excess temperature, certificate expiry, or failure of a registered control.

**P4 — fabrication controller.** The processor ranks a set of layered materials and sends one candidate to a deposition tool. In-line diffraction or spectroscopy is compared with the certified structure and curvature envelope. A mismatch updates only the calibrated branch and triggers a new certificate before the next deposition run.

**P5 — therapeutic planning.** The processor generates a source-array program for a target-region molecular or material signature. A phantom and independent dose monitor are used before any subject operation. Motion, temperature, exposure, and off-target sensors can independently terminate delivery. No clinical result is represented as achieved.

**P6 — prospective omitted-direction and local-geometry certification.** In a nondimensional synthetic two-direction qualification book, set  $A_{\text{ref}}$  to the matrix with rows (0,0.02) and (0.02,1), seed = (1,0), and initially retain only the seed. The within-retained-book residual will be zero, while the full-reference residual will be 0.02. The analytically known lower eigenvalue,  $(1 - \sqrt{1.0016})/2$ , will provide a verification target for the residual interval; these are illustrative qualification constants, not molecular

observations. The action-generated residual supplies the second direction without a selected molecular channel. In a separate adverse book  $\text{diag}(0,-1)$ , the initial seed will be exactly invariant but omit a lower state; the system will reject an unrestricted ground-state designation despite zero coupling. For a local geometry test, illustrative verified quantities  $m = 0.5$ ,  $g = 0.0001$ , and  $e_g = 0.0001$  in consistent normalized units would give a 0.0004 coordinate bound only if the basin, existence, and curvature conditions above are established. A fabrication placement command requiring 0.0003 accuracy will be withheld; one allowing 0.001 may proceed only after equipment uncertainty and all other required witnesses are included. These are prospective qualification and control examples, not claims of completed molecular accuracy or apparatus operation.

## CLAIMS

1. A configuration-ledger molecular control system comprising: an input interface configured to receive endpoint records and an integer occupancy census for a proposed molecular or material configuration; a primitive-record store holding one-entry posting records, an overlap book, and a four-index primitive pair-posting tensor; a configuration assembler configured to generate a finite many-entry configuration book from the one-entry posting records and oriented cofactors of the primitive pair-posting tensor while excluding a configuration having a duplicated orientation-and-sense slot; a stationary-ledger processor configured to determine at least one stationary configuration for which no admissible integer re-filing lowers a selected ledger value while stored closure constraints remain satisfied; a certificate engine configured to issue a machine-readable closure certificate and a convergence certificate; a control-program generator configured to convert a certified stationary configuration or a certified permitted re-filing into apparatus setpoints; and an apparatus interface configured to deliver the apparatus setpoints to physical measurement, synthesis, fabrication, or energy-delivery equipment only while the certificates satisfy stored validity criteria.
2. The system of claim 1, wherein the overlap book is a Gram matrix quantifying common filing among a plurality of charts describing a physical circulation entry, and the stationary-ledger processor solves a generalized relation  $Bc = \beta Gc$  after removing a null or near-null duplicate direction of  $G$ .
3. The system of claim 1, wherein the configuration assembler uses an exterior-cofactor lift in which pair-only additivity fixes a sign and cofactor contribution of each primitive pair-posting tensor entry to the many-entry configuration book.
4. The system of claim 1, further comprising a deterministic closure-ladder engine configured to enlarge an admissible re-filing chart according to a rule fixed without use of an observed geometry, spectrum, reaction rate, or product yield of the proposed configuration.
5. The system of claim 4, wherein the deterministic closure-ladder engine generates a block action-closure space containing  $C$ ,  $(G^{-1} B)C$ , and successive applications of  $G^{-1} B$  to  $C$  using a multi-endpoint book including cross-endpoint postings, and removes Gram-duplicated residuals at each

rung.

6. The system of claim 4, wherein the convergence certificate records changes across at least two successive rungs in stationary roots and at least one of radial moments, tensor invariants, geometry, Hessian entries, or apparatus outputs.
7. The system of claim 1, wherein the certificate engine further performs a nonsingular chart transformation or a redundant-chart embedding with Gram reconciliation and issues a chart-invariance certificate only when selected invariant outputs, transported into a common metric, remain within a stored tolerance.
8. The system of claim 1, wherein the stationary-ledger processor returns a non-closing output identifying at least one of an integer-census mismatch, failed two-ended posting, duplicate slot, impossible Gram rank, nonpositive weighted Gram book, unclosed phase registry, or unconverged closure ladder.
9. The system of claim 1, wherein the primitive-record store holds separate direct-native, calibrated, and conventional-translation branches with separate provenance and prevents a calibrated or translation update from overwriting the direct-native branch.
10. A method of controlling physical apparatus using a configuration ledger, comprising: receiving endpoint records, integer occupancies, a target, and apparatus constraints; constructing an overlap book that records duplicated filing among endpoint charts; obtaining one-entry posting records and a primitive four-index pair-posting tensor; assembling a many-entry configuration book by pair-only additive exterior cofactors while rejecting duplicate orientation-and-sense slots; testing stationary configurations against admissible integer re-filings; enlarging the chart by a predetermined closure ladder until stored convergence thresholds pass or a stop rule is reached; issuing a closure, convergence, uncertainty, and validity record; generating physical apparatus setpoints from a valid result; inhibiting the setpoints when the result is invalid or non-closing; operating the physical apparatus with the setpoints; measuring a physical output; and storing the measured output with the configuration, setpoints, certificate, and provenance.
11. The method of claim 10, further comprising rebuilding a ledger Hessian about the stationary configuration, combining a resulting curvature with an endpoint or collective inertia record, and outputting an absolute mode rate with propagated uncertainty.
12. The method of claim 10, further comprising applying a finite moving-eye correction by allowing endpoint position, inertia, registry, or motion to participate in successive ledger evaluations and reporting an order and magnitude of the correction.
13. A spectroscopic measurement and inverse-design system comprising: a configuration-ledger processor configured to assemble a finite configuration book from endpoint census records, an overlap book, one-entry posting records, and primitive pair-posting records, to reject duplicated or non-closing configurations, and to issue a convergence-certified stationary configuration or permitted whole-entry change; a controllable source; a specimen holder; an environmental

controller; a phase- and amplitude-sensitive detector; a timing reference; and a scan controller configured to generate a bounded scan or pulse program from the certified stationary configuration or permitted whole-entry change, to interleave drive-off, detuned, reference-specimen, or source-reversal controls, to compare a measured response with a certified mode or registry output, and to withhold assignment of the response when a validity witness fails.

14. The system of claim 13, wherein the configuration-ledger processor receives a target spectral response, searches candidate compositions or configurations, rejects a candidate having failed convergence or output sensitivity exceeding a stored bound, and exports a certified candidate to a synthesis or fabrication system.
15. The system of claim 13, wherein a reconciliation engine separately reports a residual attributed to a stationary configuration and a residual attributed to a preparation record, and updates only a declared calibrated primitive record or preparation field with retained prior provenance.
16. A reaction or molecular-assembly control system comprising: a configuration-ledger processor configured to assemble reactant, intermediate or saddle, and product configuration books from endpoint census, overlap, one-entry posting, and primitive pair-posting records and to issue a certified whole-entry path; a reaction vessel or molecular-assembly stage; at least one controllable thermal, pressure, flow, acoustic, optical, electrical, magnetic, mechanical, or chemical-potential source; at least one condition sensor; a product analyzer; and a reaction controller configured to reject a path requiring fractional persistent occupancy or duplicate filing, to execute a bounded apparatus program derived from a valid path and a preparation record, and to terminate or modify the program responsive to a certificate failure or off-target product measurement.
17. The system of claim 16, wherein the reaction controller certifies an event-time or yield output only when the preparation record specifies initial composition, geometry, temperature, pressure, motion, source registry, pulse history, boundary condition, and associated uncertainty adequate for the output.
18. The system of claim 16, wherein closed-loop operation changes a preparation parameter within a stored validity envelope while leaving a direct-native primitive ledger unchanged, and records measured product yield and selectivity separately from calculated stationary and path quantities.
19. A materials design and fabrication system comprising: a configuration-ledger processor configured to generate convergence-certified stationary records for candidate finite clusters, lattices, surfaces, interfaces, defects, dopants, or layered structures from endpoint census, overlap, one-entry posting, and primitive pair-posting records; a candidate generator configured to generate the candidates; a ranking engine configured to rank the candidates using certified stationary dimensions, curvatures, registry classes, barriers, and sensitivity; a fabrication interface configured to convert a selected valid candidate into deposition, lithography, implantation, annealing, mixing, printing, machining, or crystal-growth commands; a fabrication tool receiving the commands; a metrology subsystem

measuring a resulting product; and a reconciliation engine configured to compare the measured product with the certified candidate and require a new certificate before a subsequent irreversible fabrication command following an out-of-envelope mismatch.

20. The system of claim 19, wherein the ranking engine rejects a candidate when a fabrication tolerance exceeds a certified stability margin of the candidate.
21. The system of claim 19, wherein the candidate generator and ranking engine perform catalyst-site or energy-conversion-material screening and a robotic test station measures yield, selectivity, durability, storage, or conversion output for a selected candidate.
22. A diagnostic or therapeutic energy-delivery system comprising: a configuration-ledger processor configured to generate a convergence-certified target configuration or registry band from endpoint census, overlap, one-entry posting, and primitive pair-posting records; a subject or specimen model including a target-region record and at least one off-target record; a calibrated source array; at least one response, motion, temperature, or exposure sensor; an independent exposure monitor; a controller configured to generate a spatially and temporally bounded source program from the certified target configuration or registry band and to update or select successive certified control segments from sensor data; and an interlock configured independently of the controller to terminate source delivery on an exposure limit, identity failure, calibration failure, convergence-certificate failure, motion-limit failure, or off-target response.
23. The system of claim 22, wherein the calibrated source array comprises acoustic, optical, thermal, electrical, magnetic, mechanical, or particle-beam equipment and the controller stores equipment-specific validity envelopes without representing an equipment label as a primitive molecular entity.
24. The system of claim 22, further comprising a phantom qualification mode in which the source program and independent exposure monitor are tested before subject operation and a qualification record is cryptographically associated with the source program.
25. A catalyst or energy-conversion screening system comprising: a configuration-ledger processor configured to assemble and certify finite configuration books from endpoint census, overlap, one-entry posting, and primitive pair-posting records; a candidate enumerator configured to generate catalyst sites, adsorbate configurations, reaction networks, or energy-conversion materials; a certificate-aware acquisition engine configured to penalize or exclude a candidate having an unconverged, extrapolative, or non-closing ledger result; a robotic preparation station; a physical test station; and a controller configured to select a candidate, cause the stations to prepare and test the candidate, record a measured yield, selectivity, durability, or conversion output, and update a calibrated primitive-record branch while preserving a direct-native branch and prior provenance.
26. The system of claim 25, wherein the acquisition engine selects a next candidate based on both predicted performance and a certificate-derived uncertainty or sensitivity measure.

27. A non-transitory machine-readable medium storing instructions that, when executed by one or more processors coupled to physical apparatus, cause the processors to: receive endpoint and integer-census records; construct an overlap book and a primitive pair-posting tensor; assemble a finite configuration book using pair-only additive exterior cofactors; reject a duplicated or non-closing configuration; apply a predetermined chart-enlargement ladder; determine a stationary configuration and a permitted whole-entry change; issue convergence and validity certificates; generate a physical control program; and inhibit the physical apparatus from executing the control program when a required certificate fails.
28. The medium of claim 27, wherein the instructions further cause the processors to store a digital thread linking primitive-record checksums, representation rungs, stationary roots, apparatus setpoints, measured outputs, uncertainties, overrides, and validity-state transitions.
29. A method of qualifying a configuration-ledger molecular control system, comprising: loading reference endpoint and primitive pair-posting records; executing duplicate and near-duplicate Gram controls; executing chart rotation, reorder, rescaling, and overcomplete-chart controls; executing deterministic-ladder convergence tests; executing separated-entry and impossible-closure controls; evaluating at least one held-out physical reference not used to select a representation rung; commanding a dry run of connected physical apparatus; testing a safe-state interlock; storing thresholds, results, software identifiers, primitive-library identifiers, apparatus-calibration identifiers, and expiration conditions in a qualification record; and preventing issuance of a valid certificate outside the qualification record's validity envelope.
30. The method of claim 29, further comprising deliberately substituting a same-size arbitrary chart enlargement that leaves a known action-generated direction unrepresented and requiring an adverse-control failure when coupling to that direction exceeds a stored threshold.
31. The system of claim 1, wherein the certificate engine stores a certificate scope record identifying a reference generator, configuration approximation, retained and omitted effects, state sector, requested observable, and validity region; distinguishes numerical convergence, a conditional model-relative bound, and an exact-reduction certificate; retains empirical validation separately; and prevents the apparatus interface from accepting one certificate type as another.
32. The system of claim 31, wherein the certificate engine constructs an orthogonal retained-space projector in Gram-reconciled coordinates, evaluates a residual coupling the retained space to a declared complementary space under the reference generator, records unresolved-tail status, and withholds a completeness designation based solely on the projector attaining full rank in a supplied finite chart.
33. The system of claim 32, wherein the certificate engine issues the exact-reduction certificate only upon verification of retained-space invariance and containment of the requested state or prepared-state sector, and requires an omitted-sector ordering or lower-bound witness before identifying a retained state as a ground state of the reference generator.

34. The system of claim 31, wherein the certificate engine forms an output enclosure from a verified residual or complementary-block bound and observable-specific sensitivity, curvature, inertia, or branch-isolation evidence; marks an unresolved contribution as unknown; and the apparatus interface inhibits a command requiring an enclosure narrower than that supported by the evidence.
35. The system of claim 31, further comprising an adaptive refinement scheduler configured to select a predetermined representation, joint-configuration, integration, precision, or endpoint-motion refinement according to a failing certificate component and a compute budget, recompute the certificate after refinement, and return an unresolved or degraded state on exhaustion of the budget without relaxing an apparatus acceptance tolerance.
36. The method of claim 12, further comprising evaluating metric or chart derivatives and coupling associated with a position- or time-dependent retained-space projector, and invalidating or recomputing a certificate on a rank change, branch crossing, or unresolved dynamic coupling before a further apparatus command relying on that certificate.

## **ABSTRACT**

Disclosed are configuration-ledger molecular design and control systems. A processor receives endpoint records and integer occupancies, constructs overlap and primitive pair-posting records, assembles a configuration book, rejects duplicated filing, and determines stationary configurations and permitted whole-entry changes. A target-independent closure ladder generates admissible directions. A certificate engine distinguishes finite-model numerical convergence, conditional model-relative error bounds, exact reductions when supported, and empirical validation. It records omitted-direction coupling, approximation scope, observable-specific uncertainty, and provenance. Adaptive refinement targets failed certificate components; unresolved error is not treated as zero. A physical interlock accepts only certificates matching a command's accuracy and validity requirements. Apparatus embodiments include spectroscopy, reaction and synthesis, materials fabrication, catalyst screening, and diagnostic or therapeutic energy delivery. Native, calibrated, and conventional-translation records remain separate. Non-closure, numerical failure, and insufficient assurance produce distinct outputs rather than a forced physical command.

## **SCOPE AND DISTINCTIONS**

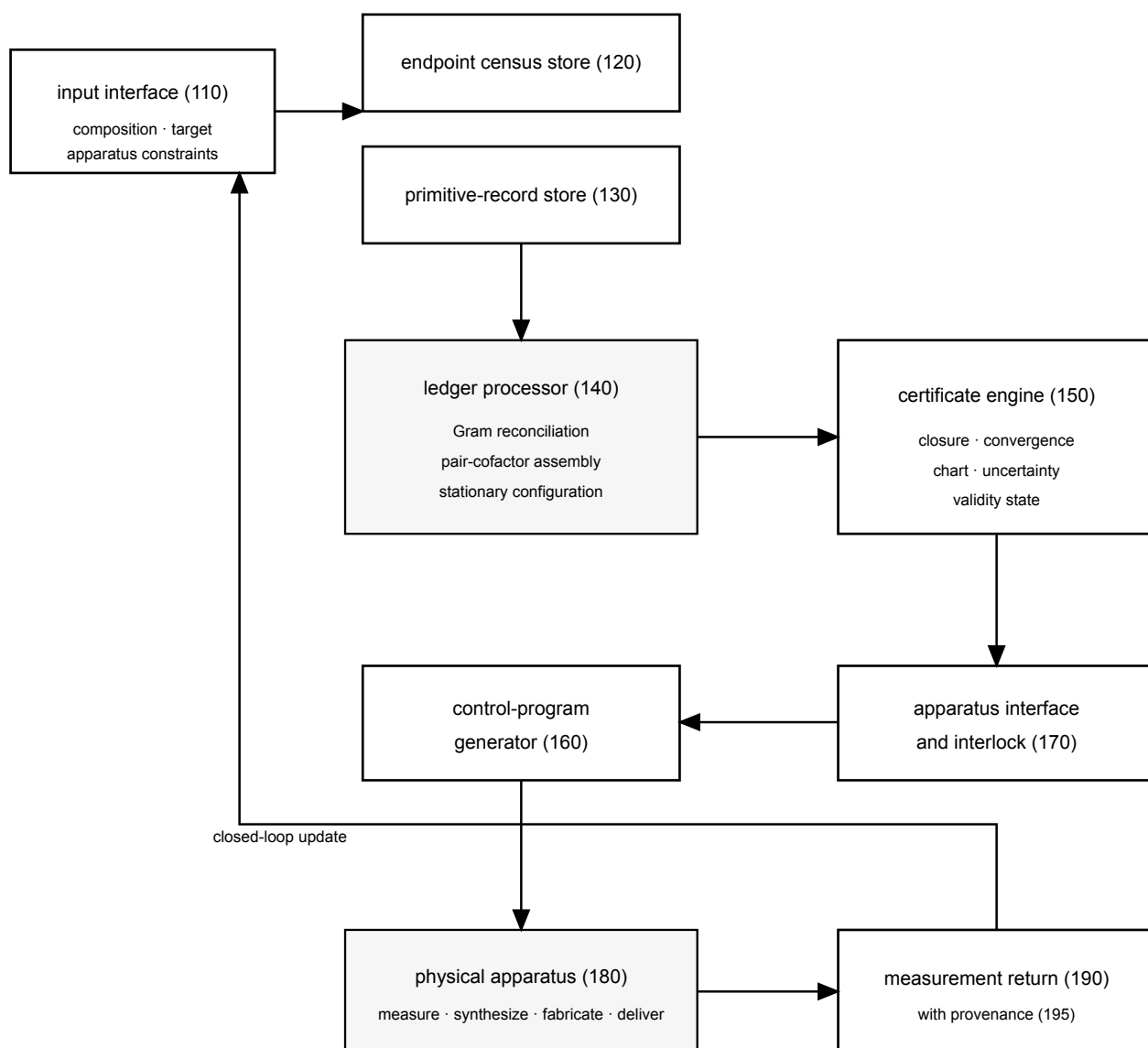
Applicant recognizes that molecular simulation, generalized matrix solvers, overlap matrices, four-index integral tensors, automated reaction screening, inverse spectroscopy, treatment planning, and closed-loop fabrication each have substantial prior art. No claim is intended to monopolize any such category, a generalized eigenvalue calculation, or a scientific law in the abstract.

The disclosed combination is directed instead to the recited operational architecture: integer endpoint census; one-entry and primitive pair-posting records; overlap-based duplicate-filing rejection; pair-only additive exterior assembly; a target-independent deterministic closure ladder; affirmative non-closure

witnesses; branch-separated native, calibrated, and translation records; machine-readable convergence and chart-invariance certificates; and a certificate interlock controlling physical apparatus. The claims should be construed as requiring their stated combination and physical use.

Medical and energy-delivery embodiments are further limited by target and off-target records, calibrated source hardware, independent exposure monitoring, and certificate-triggered termination. They do not claim treatment planning or energy delivery generally and make no assertion of clinical efficacy.

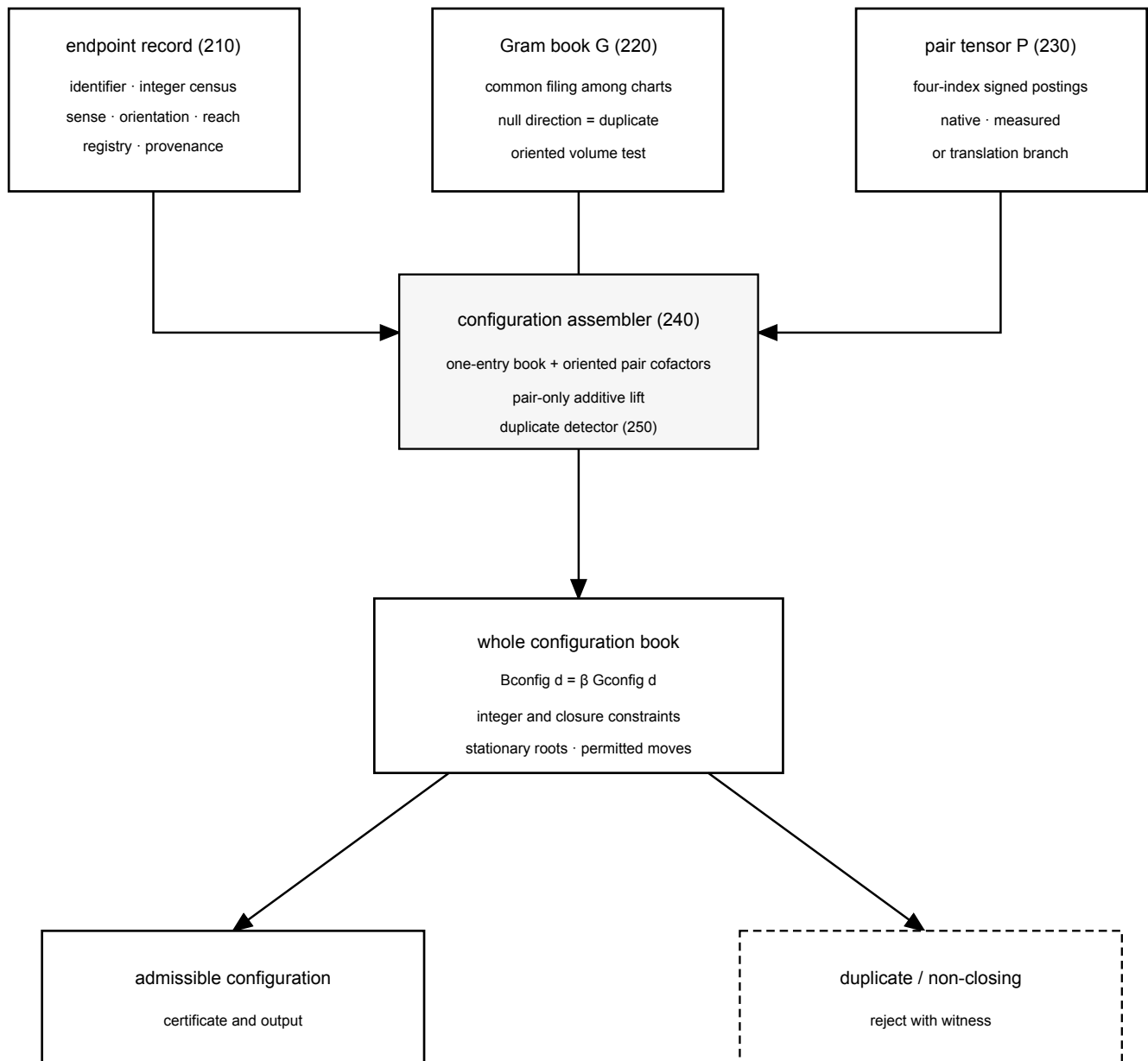
## Configuration-ledger molecular control system (100)



**FIG. 1**

Configuration-ledger processor, certificate-gated control program, physical apparatus, and provenance-preserving measurement return.

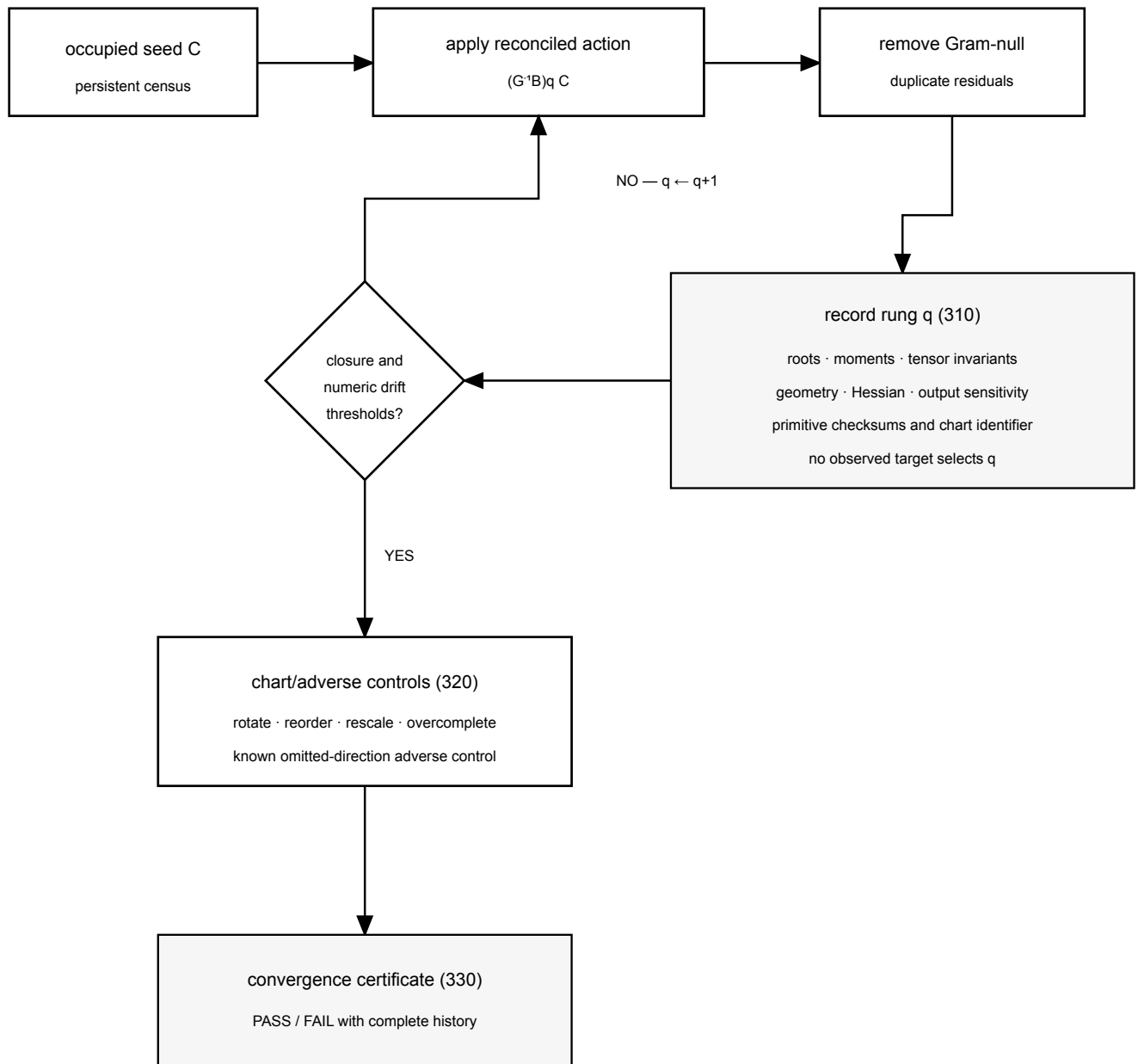
## Native records and configuration assembly (200)



**FIG. 2**

Endpoint census, overlap book, primitive pair-posting tensor, exterior-cofactor assembly, and duplicate-rejection branch.

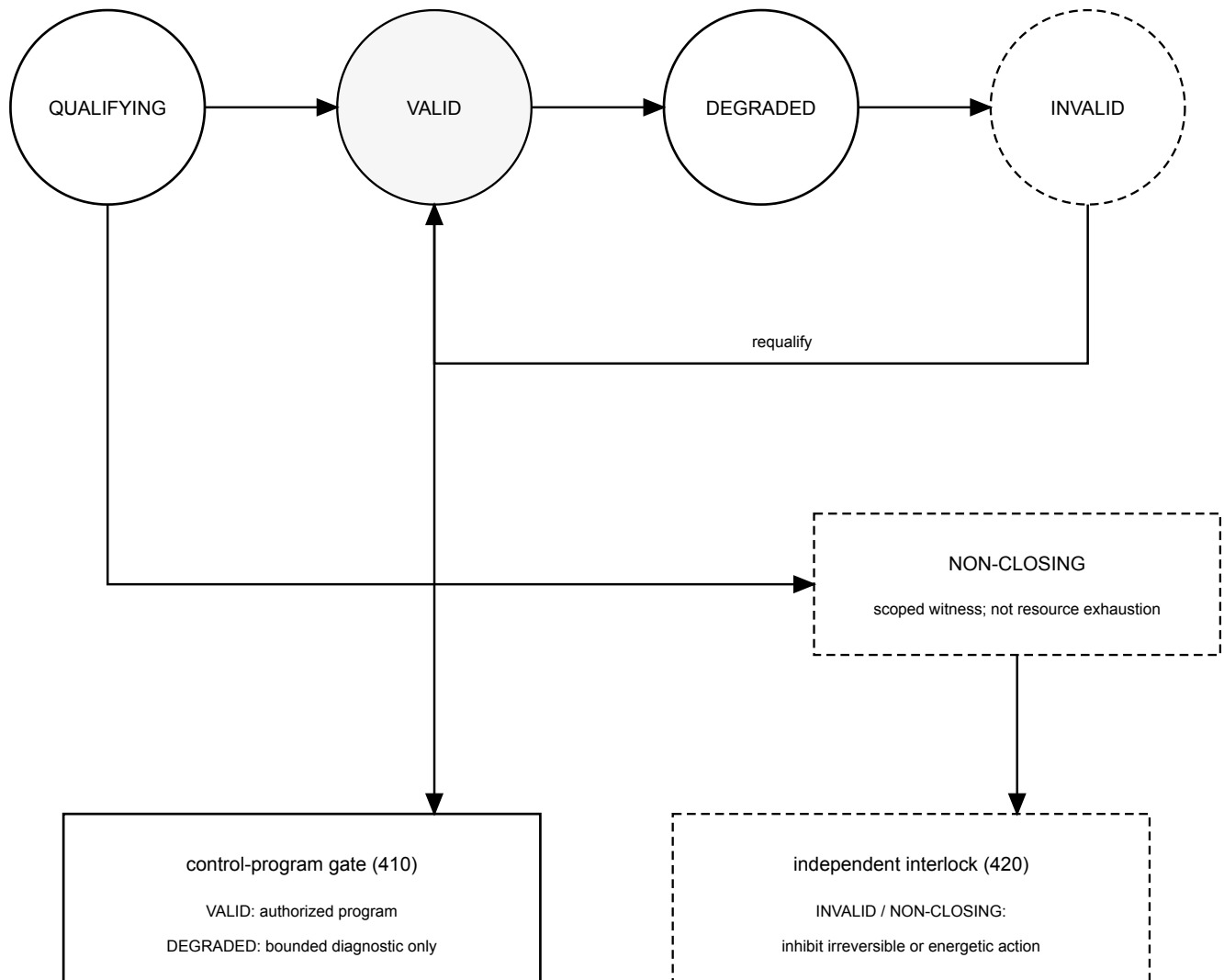
### Deterministic re-filing and convergence ladder (300)



**FIG. 3**

Predetermined closure ladder, duplicate removal, invariant closure and successive-level drift tests, chart controls, and convergence certificate.

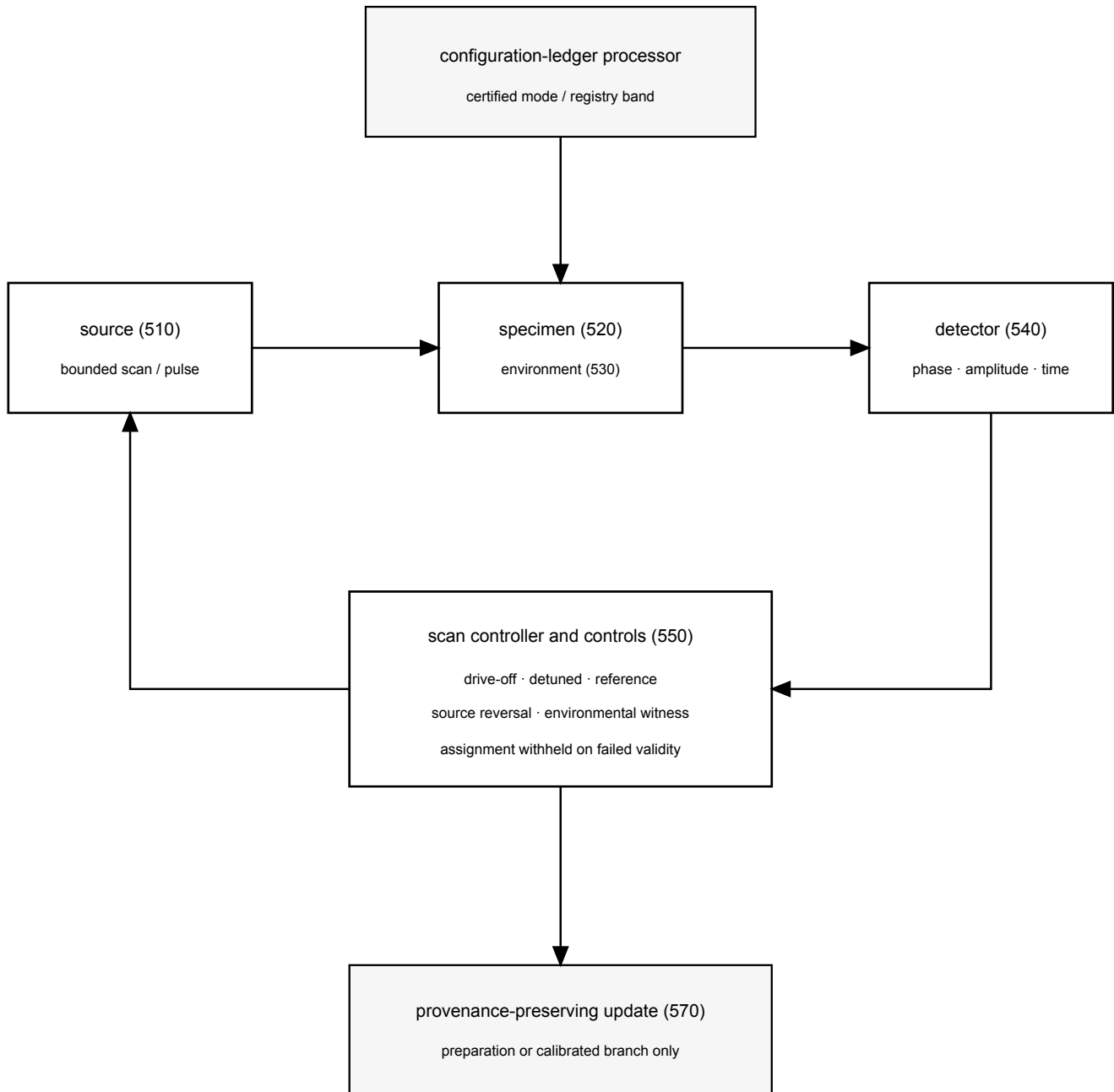
## Validity state machine and interlock (400)



**FIG. 4**

Qualification, valid, degraded, invalid, and non-closing states coupled to a physical-action interlock.

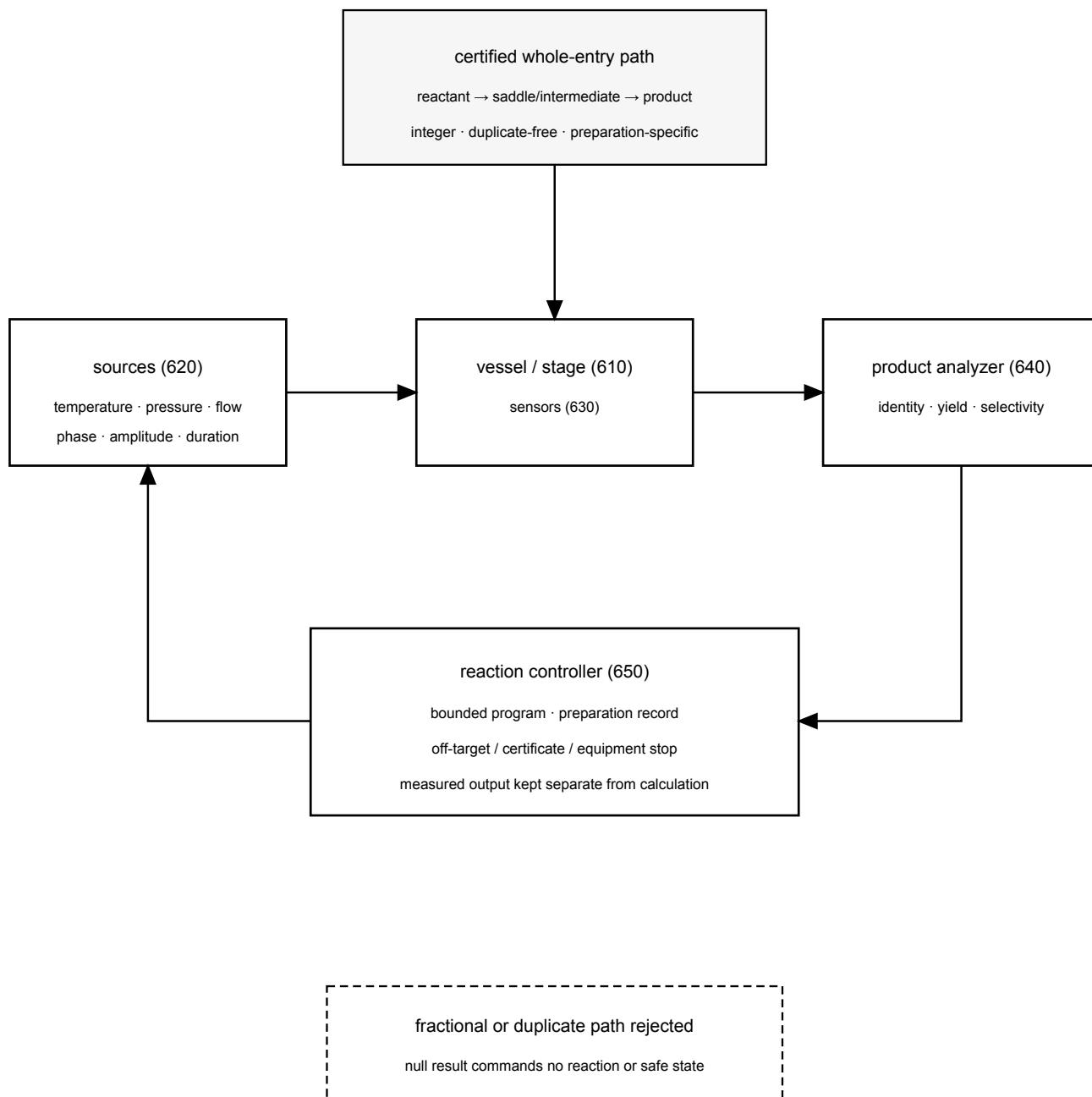
## Spectroscopic measurement and inverse design (500)



**FIG. 5**

Certified spectral program, randomized controls, phase-sensitive measurement, and branch-preserving update.

## Reaction, synthesis, and molecular assembly (600)



**FIG. 6**

Preparation-specific whole-entry pathway coupled to a bounded reaction program and product-feedback loop.

Materials design and fabrication (700)

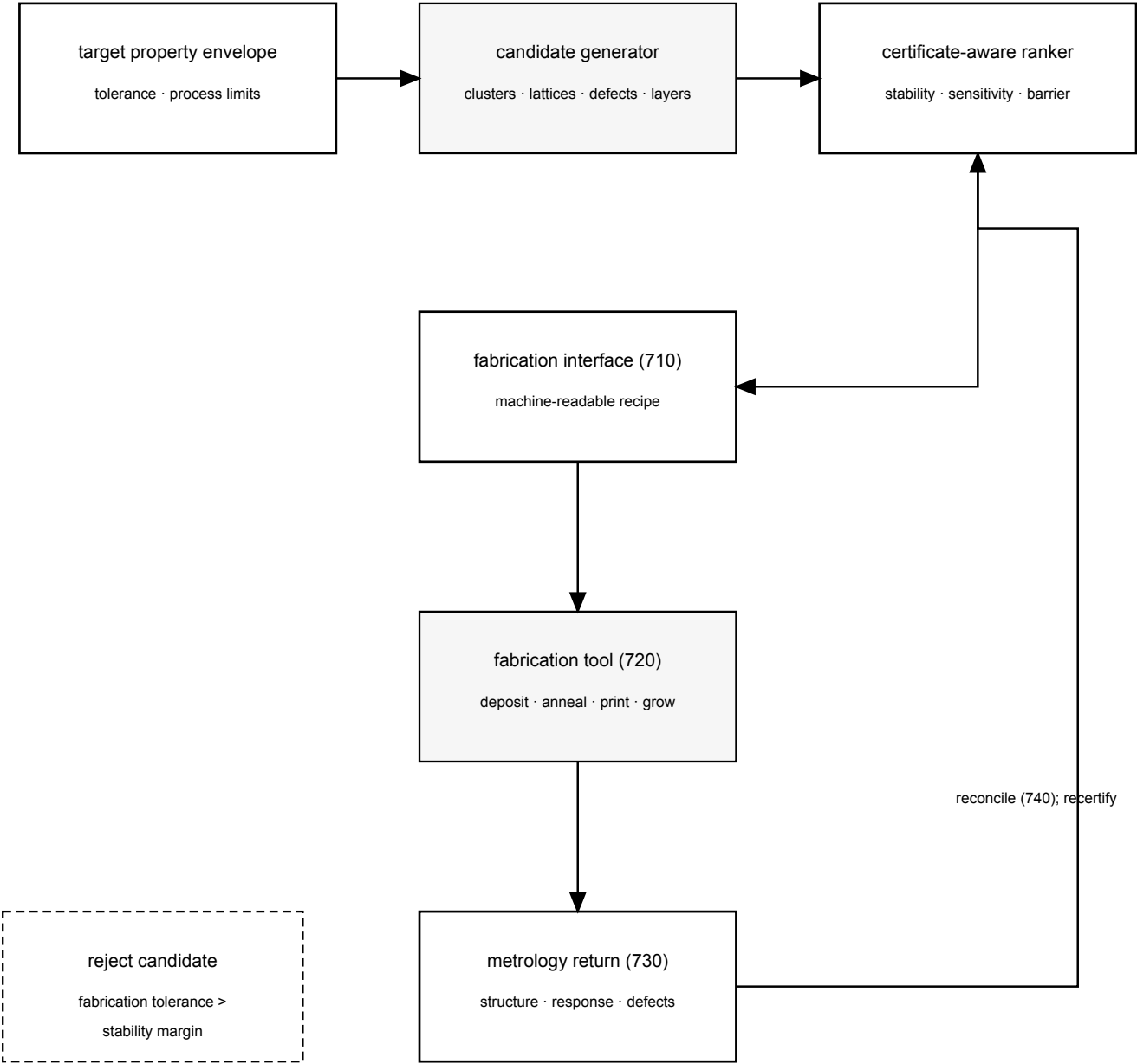


FIG. 7

Certificate-aware candidate generation, fabrication command, metrology return, and mandatory recertification.

## Diagnostic or therapeutic energy delivery (800)



**FIG. 8**

Certified target and off-target planning with bounded source array, real-time monitoring, and independent termination interlock.

Catalyst and energy-conversion screening (900)

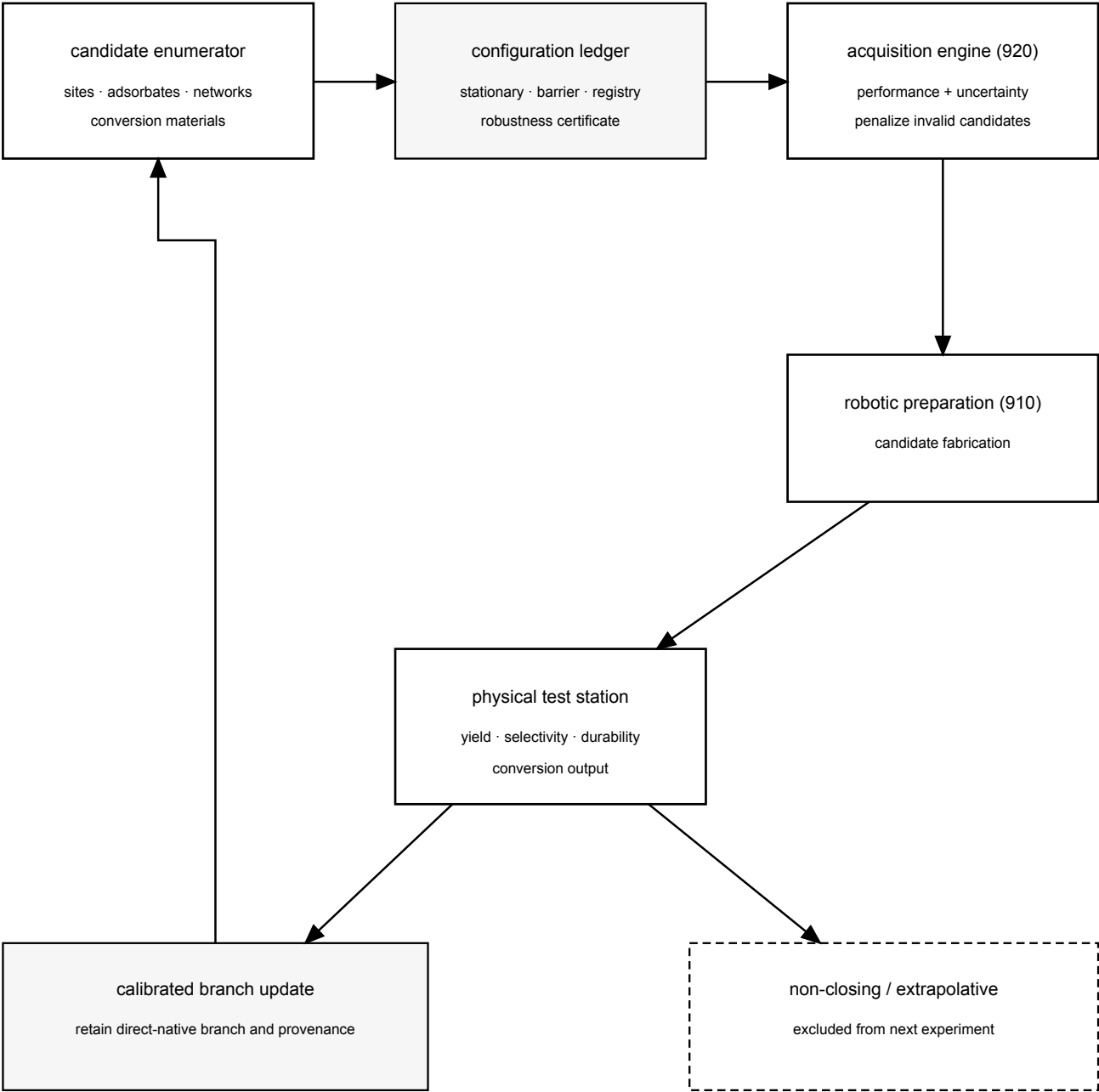


FIG. 9

Certificate-aware active screening with robotic preparation, physical testing, and provenance-preserving calibrated updates.

Qualification, digital thread, and layered scope (1000)

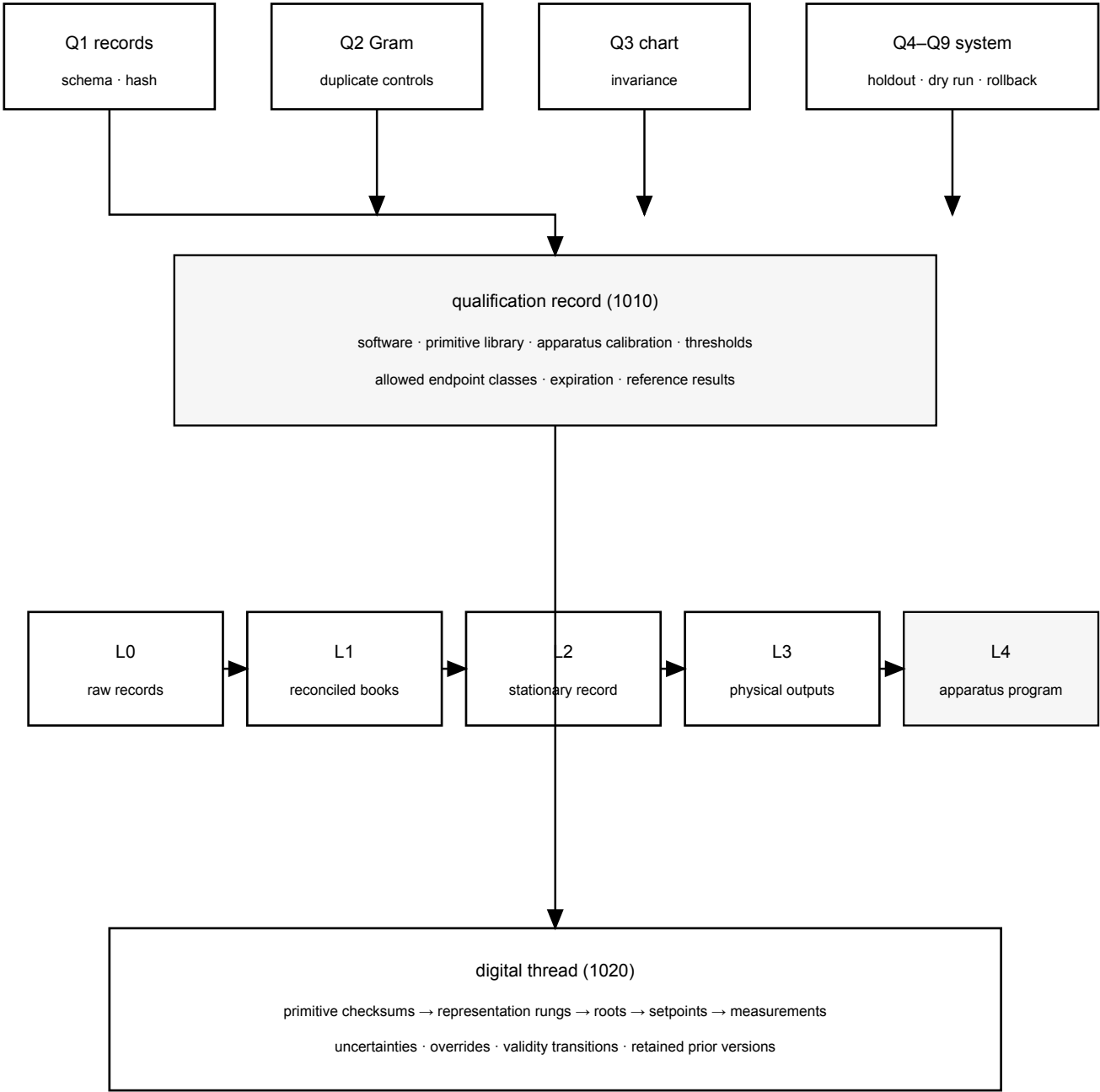
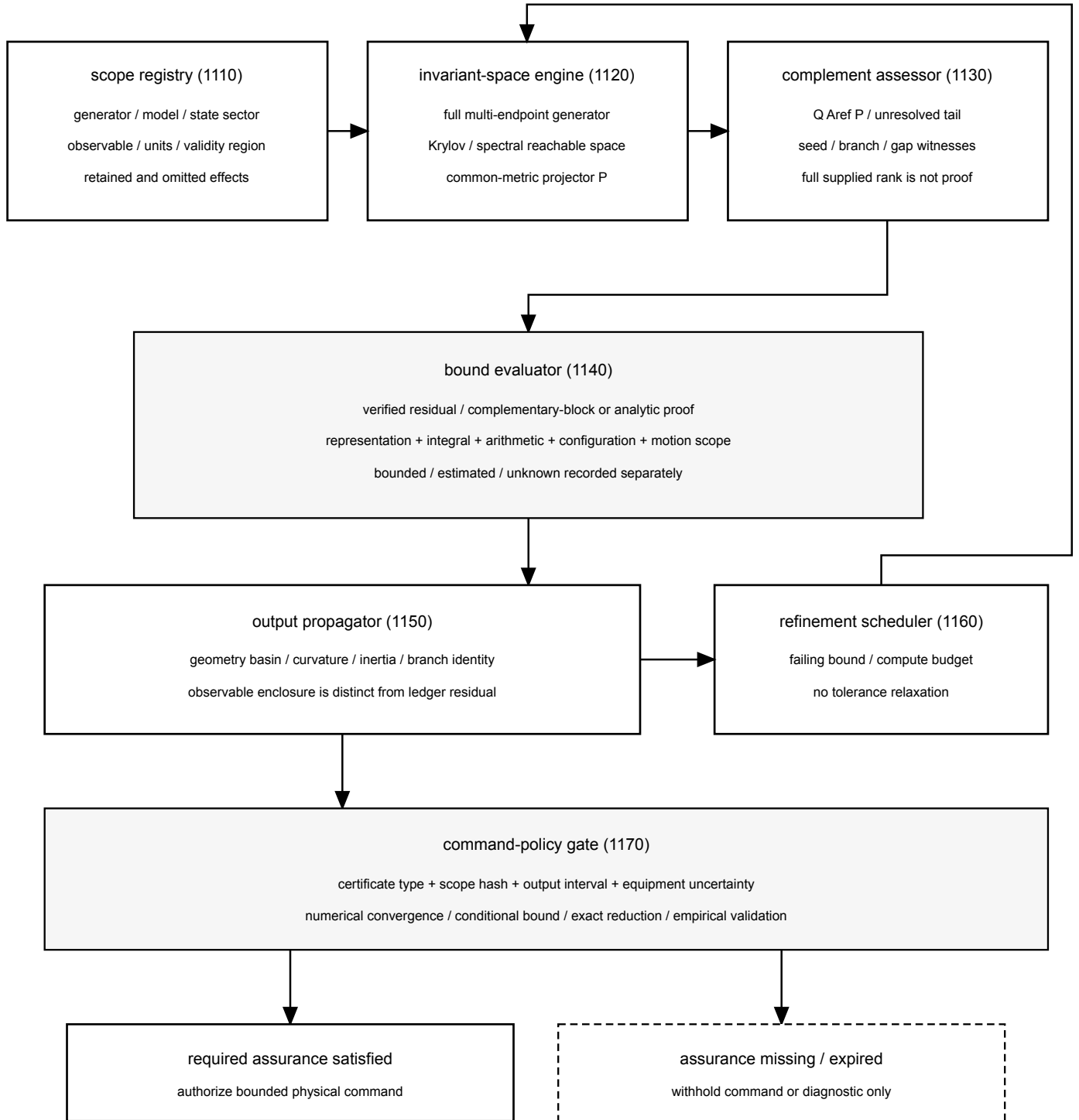


FIG. 10

Factory qualification, layered output scope, and digital thread tying primitive records to physical results and validity transitions.

## Scope-aware certification and control (1100)



**FIG. 11**

Conditional bounds and exact-reduction evidence carry declared scope through adaptive refinement to an observable-specific apparatus interlock; unknown error does not become zero.